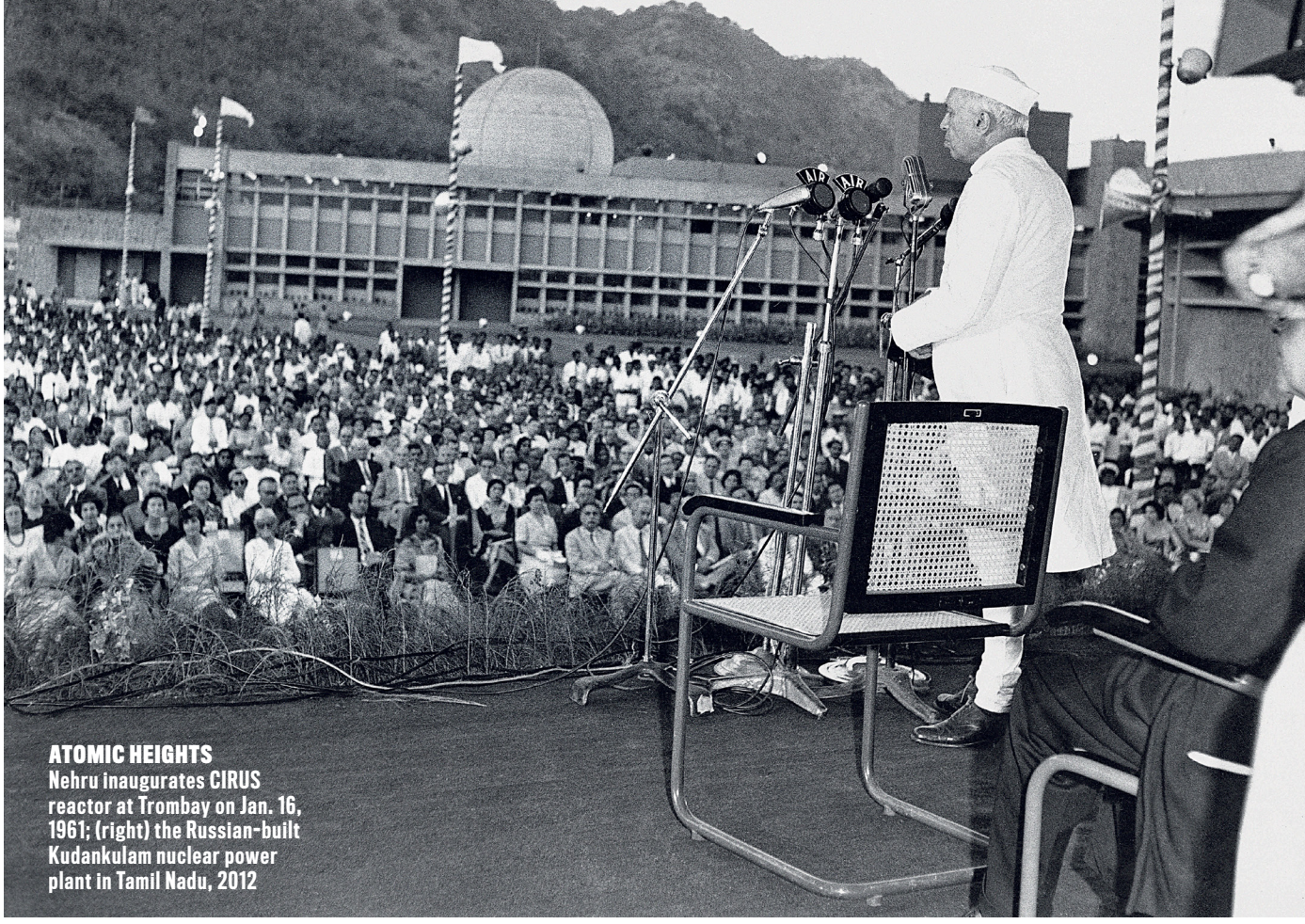


**ATOMIC HEIGHTS**
Nehru inaugurates CIRUS reactor at Trombay on Jan. 16, 1961; (right) the Russian-built Kudankulam nuclear power plant in Tamil Nadu, 2012

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# FISSION FEATS

**India's atomic journey has enabled nuclear deterrence** but failed to deliver on the promise of an energy revolution

*By* ***Amarnath K. Menon***

***India made an early commitment to nuclear power*** by enacting the Atomic Energy Act and setting up the Atomic Energy Commission (AEC) in 1948 with Homi J. Bhabha as chairman. Six years later, the Department of Atomic Energy (DAE) was created under the direct charge of Prime Minister Jawaharlal Nehru to work towards ensuring nuclear security as well as generation of electricity from the thorium and uranium available in India.

DAE pioneered research in atomic energy at Trombay, on Mumbai's outskirts, where the earliest research reactors were built. By 1963, India had two research reactors and four nuclear reactors. The DAE's robust institutional framework and its scientists' diligence enabled the resounding success of the nuclear tests at Pokhran in 1974 and 1998.

As India's uranium reserves are rather modest, while its thorium reserves are one of the world's largest, a three-stage nuclear fuel cycle strategy was adopted—natural uranium reactors, followed by fast breeder reactors using plutonium from the first generation reactors, and then reactors based on the self-sustaining thorium-uranium-233 cycle. Driven by its 'Atoms for Peace' motto, DAE focuses on delivering the benefits of nuclear science for energy, food, water, health and national security. The target of generating 10,000 MW of nuclear power by 2000, however, remains elusive even 22 years later. ■